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Can Tobacco Control Policies Prevent Tooth Loss? A Difference-in-Differences Study

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- Smoking is a modifiable cause of periodontitis and tooth loss
- Cigarette taxes and prices reduce smoking; the evidence for smoke-free laws is weaker
- Evidence linking major tobacco-control policies to oral health outcomes at population scale is limited

Primary aim (substantive)

Estimate the effect of U.S. state tobacco-control policies on smoking and on tooth loss

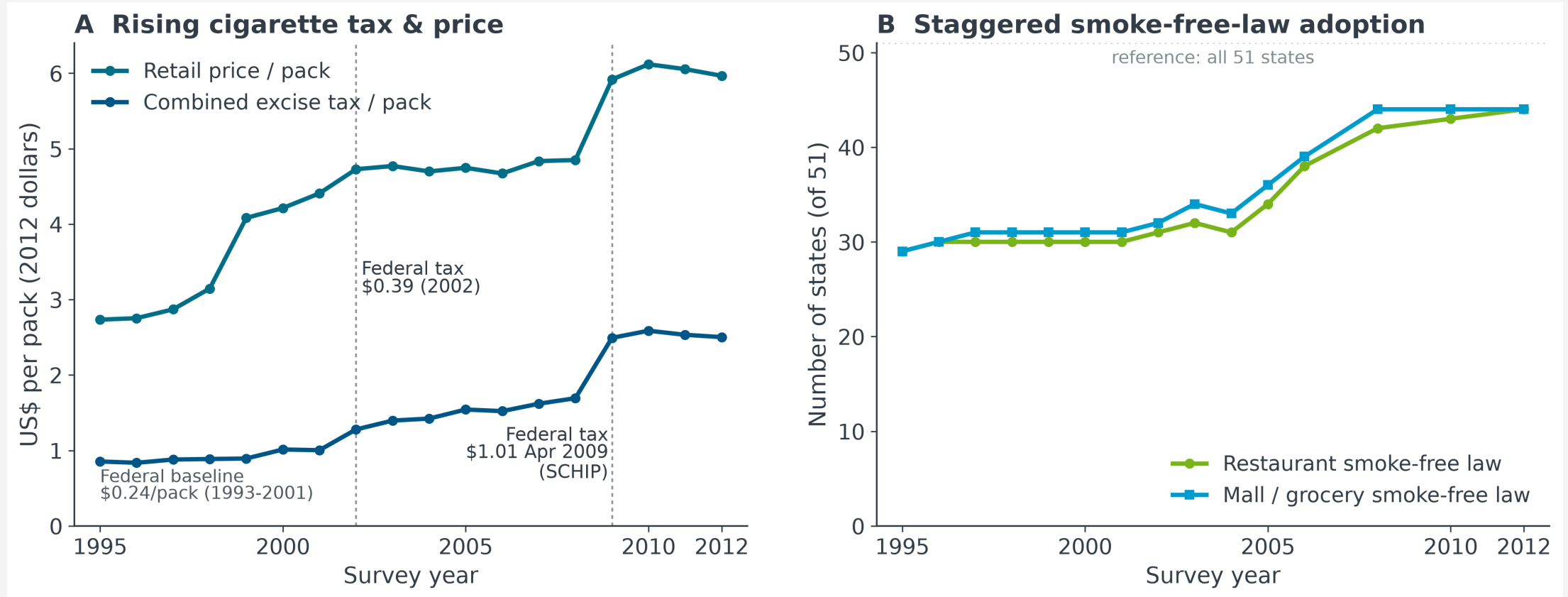
Secondary aim (methodological)

Staggered binary and continuous treatments: Compare two-way fixed effects with heterogeneity-robust DiD estimators

	Operationalisation
Sample	Behavioral Risk Factor Surveillance System (BRFSS), 1995–2012; 15 cycles, 51 states; analytic N = 2,622,010 (smoking) and 2,638,355 (tooth loss)
Treatments	Cigarette tax and price (\$/pack, continuous); restaurant and mall/grocery smoke-free laws (staggered)
Outcomes	Complete edentulism; loss of ≥ 6 teeth
Control outcomes	Positive: current smoking, quitting; negative: seat-belt use, smoke-detector

U.S. state tobacco-control policies, 1995–2012

A rising cigarette tax and price (left) and smoke-free-law adoption (right) — the variation the difference-in-differences design exploits



$$Y_{ist} = \beta D_{st} + X'_{ist}\gamma + \alpha_s + \lambda_t + \varepsilon_{ist}$$

i individual, s state, t year. D = policy (dose in \$/pack, or ban vs none); X = covariates; α = state fixed effects; λ = year fixed effects. $\beta \times 100$ is in percentage points — per +\$1.00 of dose, or for ban versus no ban

Target estimand: the average treatment effect on the treated, identified under parallel trends in the mean untreated potential outcome

Staggered binary treatment — the smoke-free laws

- β is a weighted average of every observed 2×2 DiD; the weights are non-negative (Goodman-Bacon 2021), but already-treated states serve as controls — contaminated comparisons
- Rewritten over group-time effects the weights need not be non-negative (de Chaisemartin–D'Haultfœuille 2020): the sign of the estimand is not guaranteed

Continuous dose — cigarette tax and price

- Dose in constant 2012 dollars: every state's real dose moves every year — no stayers
- No untreated cells after 1996: year effects for 1997–2012 unidentified, so the secular decline is absorbed into the dose coefficient

$$ATT(d_j | d_j) - ATT(d_{j-1} | d_{j-1}) = \underbrace{ACRT(d_j | d_j) \cdot \Delta d}_{\text{causal}} + \underbrace{ATT(d_{j-1} | d_j) - ATT(d_{j-1} | d_{j-1})}_{\text{selection on dose}}$$

d_j = dose level (\$/pack), $\Delta d = d_j - d_{j-1}$. ATT = average treatment effect on the treated: $ATT(d | d')$ is the effect of dose d among states that actually received d' . $ACRT$ = average causal response on the treated: $ACRT(d_j | d')$ is their response to moving from d_{j-1} to d_j . So comparing dose levels returns that causal response plus selection on dose (Callaway et al. 2024)

Family	Comparison group	What it needs	What it gave here
Group-time ATT Callaway–Sant'Anna 2021; Sun–Abraham 2021	Not-yet-treated (CS); the last-treated cohort where no never-treated exist (SA)	No anticipation + parallel trends; the two rest on non-nested assumptions	Positive and significant on the high-tax cohort — but fails its own positive control
Imputation Borusyak–Jaravel–Spiess 2024; Gardner 2022	All untreated cells, which fit the $Y(0)$ model	Untreated cells in every period, or the year effects are unidentified	Null throughout; the most stable family on this panel
Extended TWFE / two-way Mundlak Wooldridge 2025	None — every cell enters one regression	Homogeneous effects, or a sufficiently flexible model	Null; a specification check, not a remedy

$$ATT(g, t) = E[Y_t(g) - Y_t(0) \mid G_g = 1]$$

The group-time estimand — *identified*, not defined, by the difference-in-differences contrast against a clean comparison cohort, under no anticipation and parallel trends. Continuous dose handled as high-dose cohorts (tax $\geq$ \$3, price $\geq$ \$6 per pack)

No clean comparison for the continuous dose: the baseline-deviation specification implies –3.1 pp per \$1.00 on smoking — roughly seven times the adequately identified estimate

Smoking — percentage points (95% CI) per +\$1.00/pack

Outcome	Tax, current year	Tax, 10-year average
Current smoking	−0.35 (−0.56, −0.13)	−0.47 (−0.95, 0.01)
Quitting	0.51 (0.19, 0.82)	0.07 (−0.53, 0.67)

Tooth loss — percentage points (95% CI) per +\$1.00/pack

Outcome	Tax, current year	Tax, 10-year average
Complete edentulism	0.08 (−0.14, 0.31)	0.39 (−0.12, 0.89)
Loss of ≥6 teeth	−0.19 (−0.53, 0.15)	−0.44 (−1.17, 0.30)

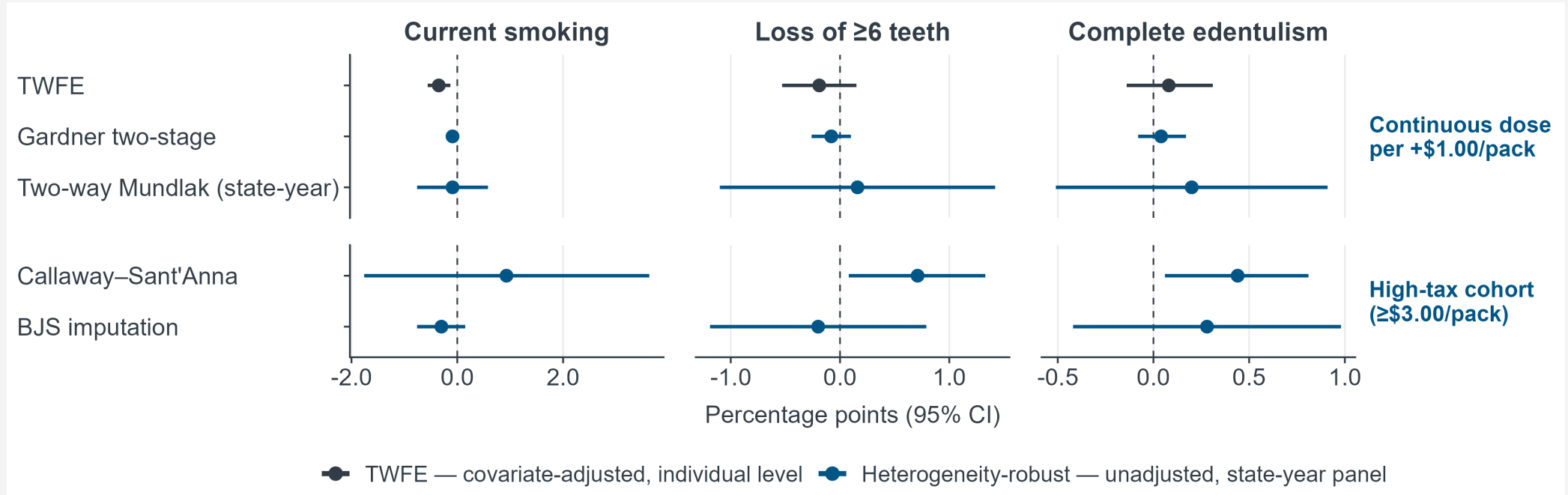
Covariate-adjusted two-way (state and year) fixed-effects estimates; standard errors clustered by state. Retail price shows similar estimates.

Percentage points (95% CI), ban vs no ban

Ban	Current smoking	Complete edentulism	Loss of ≥ 6 teeth
Restaurant	0.11 (−0.34, 0.55)	0.45 (0.03, 0.88)	0.51 (−0.02, 1.05)
Mall / grocery	0.30 (−0.21, 0.81)	0.24 (−0.19, 0.68)	0.52 (−0.15, 1.18)

- No first stage: neither ban lowered smoking, and several tooth-loss estimates are positive
- The mall/grocery ban fails its negative control: smoke-detector test −1.17 pp (−2.27, −0.08)
- The tax passes both negative controls (both null)

Estimator comparison

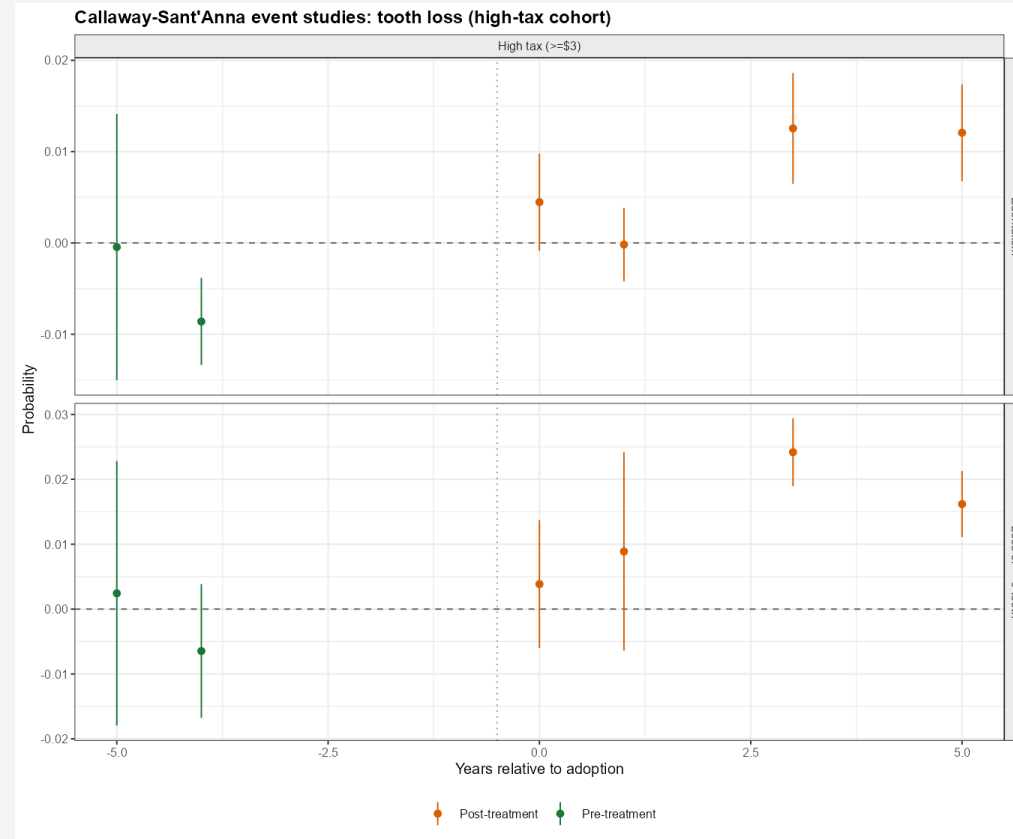


- Left panel: no robust estimator recovers the first stage on the state-year panel — but only Callaway–Sant'Anna gets the sign wrong (+0.93 pp, -1.76 to 3.63), and only it turns positive for tooth loss
- Different estimands: continuous per +\$1.00 dose versus binary $\geq \$3.00$ -cohort ATT
- Callaway–Sant'Anna rests on small, late, biennially observed cohorts; imputation, more stable, stays null

Sun–Abraham omitted here: for the high-dose tooth-loss cohorts — small, late and biennially observed — its variance is undefined (the group-time estimator does not converge there either); it is estimable for the bans

Callaway–Sant'Anna event study (Roth 2026)

Universal base period (Roth 2026). Pre-treatment coefficients are not negligible — -0.6 to -0.9 pp, larger than any effect of interest — and post-treatment coefficients turn significantly positive. Read together: a differential trend, not a protective policy



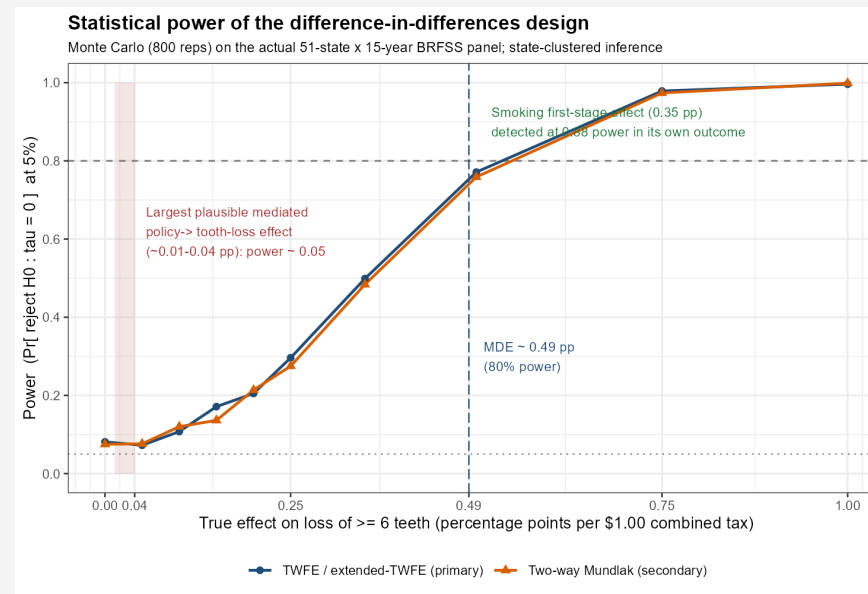
No $t-1$ (universal base period); no $t-3$ or $t-2$ (the oral-health module ran annually to 2006, then every second year); and the high-dose cohorts adopt late

$$E[\Delta Y(0) \mid D = 1] = E[\Delta Y(0) \mid D = 0]$$

- A pre-trend of the size just shown is exactly what this assumption forbids
- Untestable: pre-trend tests have low power, and conditioning on passing one distorts inference (Roth 2022) — judge magnitude
- Identification also needs a stable-composition condition; both are conditions on population means, not individual trajectories
- Rambachan–Roth (2023) relative magnitudes ($\bar{M} = 2$): the high-tax event-study interval for ≥ 6 teeth widens from $[-0.20, 1.50]$ to $[-9.0, 11.1]$ pp

Is the null informative? Minimum detectable effect and plausibility

- CI excludes protective effects beyond ≈ 0.5 pp for the current-year tax, ≈ 1.2 pp for the 10-year average — a precise null, and an equivalence statement rather than evidence of absence (Rainey 2014; Lakens et al. 2018)
- Minimum detectable effect 0.49 pp, against a plausible mediated effect of 0.01–0.04 pp — 10–40 $\times$ below it
- The ceiling is between-state variation and 51 clusters, not the 2.6 m respondents (Bertrand et al. 2004)
- Detecting even the most generous plausible effect (0.04 pp) would take $\approx 7,700$ states, not 51 — more respondents or more years do not close that gap (backup)



1. Cigarette taxes and prices reduce smoking; we find no detectable effect on tooth loss
2. That null is a limit of the design, not a finding about teeth
3. The robust estimators diverge — but that is a small-sample and parallel-trends result, not a substantive one: the estimator that turns positive fails both its positive control and its pre-trends
4. Continuous-dose difference-in-differences is the binding constraint here: no stayers, no untreated cells, and dose contrasts that carry selection on dose



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Thank you

Paper in print in the *Journal of Dental Research*

Slides and code:

github.com/BaumeisterS/DGEPI2026_tobacco_policy_tooth_loss

DiD estimators: Goodman-Bacon A. 2021. Difference-in-differences with variation in treatment timing. *J Econom.* 225(2):254–277. de Chaisemartin C, D'Haultfœuille X. 2020. Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects. *Am Econ Rev.* 110(9):2964–2996. Callaway B, Sant'Anna PHC. 2021. Difference-in-Differences with multiple time periods. *J Econom.* 225(2):200–230. Sun L, Abraham S. 2021. Estimating dynamic treatment effects in event studies with heterogeneous treatment effects. *J Econom.* 225(2):175–199. Borusyak K, Jaravel X, Spiess J. 2024. Revisiting Event-Study Designs: Robust and Efficient Estimation. *Rev Econ Stud.* 91(6):3253–3285. Gardner J. 2022. Two-stage differences in differences. *arXiv:2207.05943*. Liu L, Wang Y, Xu Y. 2024. A Practical Guide to Counterfactual Estimators for Causal Inference with Time-Series Cross-Sectional Data. *Am J Polit Sci.* 68(1):160–176. Wooldridge JM. 2025. Two-way fixed effects, the two-way Mundlak regression, and difference-in-differences estimators. *Empir Econ.* 69(5):2545–2587. de Chaisemartin C, D'Haultfœuille X, Vazquez-Bare G. 2024. Difference-in-Difference Estimators with Continuous Treatments and No Stayers. *AEA Pap Proc.* 114:610–613. Callaway B, Goodman-Bacon A, Sant'Anna PHC. 2024. Difference-in-differences with a Continuous Treatment. NBER Working Paper 32117.

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Backup: the four 2×2 comparisons under staggered adoption

TWFE averages over every 2×2 difference-in-differences the data allow. Three are clean; the fourth is not

	Treated group	Control group	Clean?
1	Early adopters	Never-treated	Yes
2	Late adopters	Never-treated	Yes
3	Late adopters	Early adopters, before they adopt	Yes
4	Early adopters	Late adopters, while the early group is already treated	No

In comparison 4 the control group is already treated, so its outcome change carries its own evolving treatment effect rather than a clean counterfactual trend. Subtracting it subtracts part of an effect — which is why the bias term is non-zero only when treatment effects change over time

Backup: what TWFE estimates under staggered adoption

The two decompositions behind slide 5 — the weights are non-negative in the first and need not be in the second

$$\hat{\beta}^{\text{TWFE}} = \sum_k s_k \widehat{\text{DD}}_k, \quad s_k \geq 0, \quad \sum_k s_k = 1$$

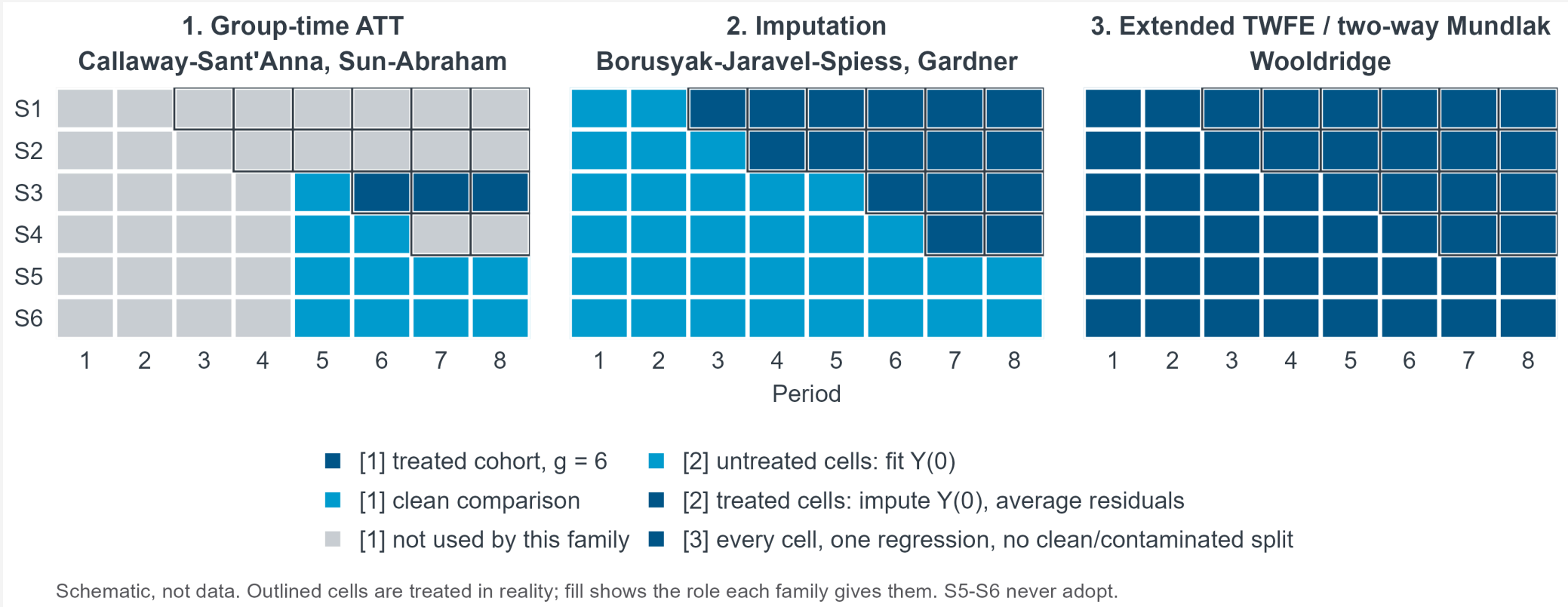
Goodman-Bacon (2021): a variance-weighted average of all observed 2×2 DiDs. The weights are variance shares, so they cannot be negative; the bias enters because some 2×2 comparisons use already-treated units as controls

$$\beta^{\text{TWFE}} = \sum_{g,t} w_{g,t} \text{ATT}(g,t), \quad \exists w_{g,t} < 0$$

de Chaisemartin–D'Haultfœuille (2020): rewritten over group-time effects, the weights need not be non-negative — so β can carry the wrong sign even when every underlying $\text{ATT}(g,t)$ has the same sign

Backup: which panel cells each family uses

The same three families, drawn on a stylised panel with staggered adoption



Backup: inference with 51 clusters

- Standard errors clustered by state throughout (51 clusters)
- Monte Carlo on the actual 51-state $\times$ 15-year panel (800 replications): power ≈ 0.50 , 0.77 and 0.98 at true effects of 0.35, 0.50 and 0.75 pp
- The simulation shows a mild finite-cluster over-rejection (size ≈ 0.08) — conservative for a no-effect conclusion
- A CR2 bias-reduced, Satterthwaite-corrected small-cluster variance leaves the intervals essentially unchanged, confirming the 51-cluster inference is not anti-conservative
- The equivalence bound is read from the interval itself, not from the simulated size

Backup: what design would detect the plausible effect?

Suppl. Table 1 gives $SE = 0.174$ pp and $MDE = 0.49$ pp for loss of ≥ 6 teeth, i.e. $MDE = 2.8 \times SE$ (80% power, 5% two-sided). Since the ceiling is between-state variation and the number of clusters, precision scales as $1/\sqrt{G}$ — so the states required are $51 \cdot (0.49/\text{target})^2$

Target effect	SE needed	States needed	vs the 51 available
0.04 pp — most generous plausible	0.014 pp	$\approx 7,700$	150×
0.013 pp — central chained estimate	0.005 pp	$\approx 72,000$	1,400×
0.01 pp — lower end	0.004 pp	$\approx 122,000$	2,400×

- More respondents do not help: the binding constraint is between-state variation, not sampling error. Were it otherwise, 0.04 pp would still need ≈ 400 million respondents
- More follow-up years buy little precision either — serially correlated state shocks are exactly what Bertrand et al. (2004) show defeats naive time-series gains; longer follow-up raises the effect to be found rather than lowering the threshold

Not formally addressed in the paper — the argument, not a published result

- Interference is plausible: residents of a high-tax state can buy across a state line, so a neighbour's consumption depends on this state's tax
- Direction: cross-border purchasing weakens the realised dose in treated states, attenuating the first stage toward the null
- The first stage is nonetheless detected (-0.35 pp per \$1.00), so attenuation does not overturn it
- For tooth loss the design is already far short of the minimum detectable effect, so interference is not what produces the null